

Electromagnetics Foundations

A Beginner-Friendly Study Guide

Vector Calculus, Electric Fields, Maxwell's Equations,
Electromagnetic Waves, Fourier Analysis, and SDR Connections

Purpose:

Build the mathematical and physical foundation needed
to understand graduate-level electromagnetics.

Boxed equations = equations worth memorizing

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1 The Big Picture

Electromagnetics is fundamentally about understanding how **electric fields** and **magnetic fields** are produced, how they interact with matter, and how they propagate through space.

A useful mental hierarchy is:

1. Scalars and vectors
2. Vector calculus
3. Electric charge and current
4. Electric and magnetic fields
5. Constitutive relations
6. Maxwell's equations
7. Electromagnetic waves
8. Fourier analysis and frequency-domain thinking

The most important conceptual chain is:

Charge/current $\rightarrow$ fields $\rightarrow$ Maxwell's equations $\rightarrow$ waves $\rightarrow$ energy/information propagation
--

For SDR work, Fourier analysis provides another important chain:

Time-domain signal $\rightarrow$ Fourier transform $\rightarrow$ frequency-domain representation $\rightarrow$ filtering/modulation/demodulation
--

2 Scalars and Vectors

A **scalar** has only magnitude.

Examples:

- Temperature
- Voltage
- Electric charge
- Frequency
- Permittivity

A **vector** has both magnitude and direction.

Examples:

- Electric field **E**
- Magnetic field **H**

- Current density $\mathbf{J}$
- Force $\mathbf{F}$
- Position $\mathbf{r}$

A three-dimensional vector can be written as

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

where:

- $\mathbf{A}$ = vector
- A_x, A_y, A_z = components of the vector
- $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ = unit vectors in the x , y , and z directions

3 Unit Vectors

A unit vector has magnitude 1.

The unit vector pointing in the direction of $\mathbf{A}$ is

$$\hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|}$$

The hat indicates that the vector has been normalized.
For example,

$$\hat{\mathbf{x}}$$

means “the unit vector pointing in the x direction.”

4 Vector Magnitude

The magnitude is the length of a vector.

For

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}},$$

the magnitude is

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

This is just the three-dimensional version of the Pythagorean theorem.

5 Dot Product

The dot product takes two vectors and produces a scalar.

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta$$

where θ is the angle between the vectors.

The dot product tells you how much one vector points in the direction of another.

Important cases:

$$\mathbf{A} \cdot \mathbf{B} > 0$$

means the vectors generally point in similar directions.

$$\mathbf{A} \cdot \mathbf{B} = 0$$

means the vectors are perpendicular.

$$\mathbf{A} \cdot \mathbf{B} < 0$$

means they generally point in opposite directions.

The component form is

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z.$$

6 Cross Product

The cross product takes two vectors and produces another vector.

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin \theta \hat{\mathbf{n}}$$

where:

- θ = angle between the vectors
- $\hat{\mathbf{n}}$ = unit vector perpendicular to both

The cross product is zero when the vectors are parallel.

It is largest when the vectors are perpendicular.

Direction is determined using the right-hand rule.

7 Scalar Fields and Vector Fields

A **scalar field** assigns a scalar value to every point in space.

Example:

$$V(x, y, z)$$

is a voltage field.

A **vector field** assigns a vector to every point in space.

Example:

$$\mathbf{E}(x, y, z)$$

is an electric-field vector field.

This distinction is extremely important because vector calculus operators behave differently depending on whether their input is a scalar or vector.

8 Partial Derivatives

A partial derivative tells you how a function changes with respect to one variable while treating the others as constant.

For

$$V(x, y, z),$$

we can calculate

$$\frac{\partial V}{\partial x}, \quad \frac{\partial V}{\partial y}, \quad \frac{\partial V}{\partial z}.$$

For example,

$$V(x, y) = x^2 + 3y,$$

gives

$$\frac{\partial V}{\partial x} = 2x$$

and

$$\frac{\partial V}{\partial y} = 3.$$

9 The Nabla Operator

The symbol

$$\nabla$$

is called **nabla** or **del**.

In Cartesian coordinates,

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}$$

Think of ∇ as a mathematical operator that performs spatial derivatives.

It can be combined with fields in several important ways:

$$\nabla V$$

gradient,

$$\nabla \cdot \mathbf{A}$$

divergence,
and

$$\nabla \times \mathbf{A}$$

curl.

10 Gradient

The gradient takes a scalar field and produces a vector.

$$\nabla V = \frac{\partial V}{\partial x} \hat{\mathbf{x}} + \frac{\partial V}{\partial y} \hat{\mathbf{y}} + \frac{\partial V}{\partial z} \hat{\mathbf{z}}$$

The gradient points in the direction where the scalar field increases most rapidly.
For voltage,

$$\mathbf{E} = -\nabla V$$

This means the electric field points toward decreasing electric potential.

11 Divergence

Divergence measures how much a vector field is “spreading out” from a point.
For

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}},$$

the divergence is

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

The result is a scalar.

Intuitively:

- Positive divergence $\rightarrow$ source
- Negative divergence $\rightarrow$ sink
- Zero divergence $\rightarrow$ no net source/sink

12 Curl

Curl measures local rotation or circulation of a vector field.

$$\nabla \times \mathbf{A}$$

For

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}},$$

the expanded expression is

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}.$$

Therefore,

$$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{\mathbf{z}}.$$

13 Laplacian

The Laplacian of a scalar field is

$$\nabla^2 V = \nabla \cdot (\nabla V)$$

In Cartesian coordinates,

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

The Laplacian tells you how a quantity compares to its surrounding values.

It appears constantly in electromagnetics, especially in electrostatics and wave equations.

14 Integrals

An integral is essentially accumulation.

A one-dimensional integral,

$$\int_a^b f(x) dx,$$

adds up infinitely many small contributions.

In three-dimensional electromagnetics, you will encounter:

$$\iiint_V$$

for volume integrals,

$$\iint_S$$

for surface integrals,
and

$$\oint_C$$

for closed line integrals.

The subscript tells you what geometric object you are integrating over.

15 Differential Displacement

The differential displacement vector represents a tiny movement through space.

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$$

where:

- dx = tiny displacement in x
- dy = tiny displacement in y
- dz = tiny displacement in z

16 Differential Surface Area

A small surface element is represented by

$$d\mathbf{S} = \hat{\mathbf{n}} dS$$

where:

- dS = tiny area
- $\hat{\mathbf{n}}$ = unit vector perpendicular to the surface
- $d\mathbf{S}$ = vector area element

17 Closed Integrals

A circle over an integral sign means the path or surface is closed.

For example,

$$\oint_C$$

means a closed line integral around contour C .

Similarly,

$$\oint_S$$

means a closed surface integral.

This notation becomes extremely important in Maxwell's equations.

18 Flux

Flux describes how much of a vector field passes through a surface.

For a uniform field,

$$\Phi = EA \cos \theta$$

where:

- Φ = flux
- E = magnitude of the field
- A = surface area
- θ = angle between the field and surface normal

The general electric-flux expression is

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{S}$$

The dot product is important because only the component of $\mathbf{E}$ perpendicular to the surface contributes to flux.

19 Charge

Electric charge is measured in coulombs.

The total charge contained inside a volume is

$$Q = \int_V \rho \, dv$$

where:

- Q = total charge
- ρ = volume charge density
- dv = differential volume

20 Charge Density

Charge can be distributed through a volume, across a surface, or along a line.

Volume charge density:

$$\rho = \frac{dQ}{dv}$$

Surface charge density:

$$\rho_s = \frac{dQ}{dS}$$

Line charge density:

$$\rho_l = \frac{dQ}{dl}$$

Units are:

$$[\rho] = \text{C/m}^3$$

$$[\rho_s] = \text{C/m}^2$$

$$[\rho_l] = \text{C/m}.$$

21 Electric Potential

Electric potential, or voltage, is measured in volts.

It represents electrical potential energy per unit charge.

The electric field and voltage are related by

$$\mathbf{E} = -\nabla V$$

The negative sign means that the electric field points toward decreasing voltage.

For a one-dimensional situation,

$$E_x = -\frac{dV}{dx}.$$

22 Electric Field

The electric field is force per unit charge.

$$\mathbf{E} = \frac{\mathbf{F}}{q}$$

where:

- $\mathbf{E}$ = electric field
- $\mathbf{F}$ = electric force
- q = charge

The units are

$$\text{N/C}$$

or equivalently

$$\text{V/m}.$$

23 Permittivity

Permittivity describes how a material responds to an electric field.

The symbol is

$$\epsilon.$$

The constitutive relation is

$$\mathbf{D} = \epsilon \mathbf{E}$$

where:

- $\mathbf{D}$ = electric flux density
- ϵ = permittivity
- $\mathbf{E}$ = electric field

Permittivity is often written as

$$\epsilon = \epsilon_r \epsilon_0$$

where:

- ϵ_r = relative permittivity
- ϵ_0 = permittivity of free space

A material with larger ϵ generally stores more electric energy for a given electric field.

24 Dielectrics

A dielectric is an electrically insulating material that can become polarized when exposed to an electric field.

Examples include:

- Glass
- Ceramic
- Plastic
- Air

The relative permittivity is

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}.$$

If

$$\epsilon_r > 1,$$

the material has greater permittivity than free space.

25 Electric Flux Density

Electric flux density is represented by

D.

For a linear isotropic material,

$$\boxed{\mathbf{D} = \epsilon \mathbf{E}}$$

The units of **D** are

$$\text{C/m}^2.$$

Do not confuse **D** with **E**:

E = electric field

while

D = electric flux density.

26 Magnetic Flux Density and Magnetic Field Intensity

There are two commonly used magnetic-field quantities:

B

and

H.

They are related through

$$\boxed{\mathbf{B} = \mu \mathbf{H}}$$

where:

- **B** = magnetic flux density
- **H** = magnetic field intensity
- μ = permeability

Similarly,

$$\boxed{\mu = \mu_r \mu_0}$$

where:

- μ_r = relative permeability
- μ_0 = permeability of free space

27 Current Density

Current density describes how electric current is distributed through an area.

It is represented by

$$\mathbf{J}.$$

For a simple ohmic material,

$$\mathbf{J} = \sigma \mathbf{E}$$

where:

- $\mathbf{J}$ = current density
- σ = electrical conductivity
- $\mathbf{E}$ = electric field

The units of $\mathbf{J}$ are

$$\text{A/m}^2.$$

28 The Three Constitutive Relations

For simple linear isotropic materials, remember:

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{B} = \mu \mathbf{H}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

These describe how a material responds to electric and magnetic fields.

29 Maxwell's Four Equations

These are the central equations of classical electromagnetics.

29.1 Gauss's Law for Electricity

$$\nabla \cdot \mathbf{D} = \rho$$

Electric charge acts as a source or sink of electric flux.

29.2 Gauss's Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0$$

There are no isolated magnetic monopoles in classical electromagnetics.

29.3 Faraday's Law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

A changing magnetic field produces a circulating electric field.

29.4 Ampere-Maxwell Law

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Electric current and changing electric fields produce circulating magnetic fields.

30 Maxwell's Equations: Physical Picture

A useful memorization table is:

Equation	Mathematical operation	Physical idea
$\nabla \cdot \mathbf{D} = \rho$	Divergence	Charge creates electric flux
$\nabla \cdot \mathbf{B} = 0$	Divergence	No magnetic monopoles
$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$	Curl	Changing magnetic field creates electric field
$\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$	Curl	Current/changing electric field creates magnetic field

The deepest idea is that changing electric and magnetic fields can sustain each other and propagate through space.

31 Poisson's Equation

Start with

$$\mathbf{E} = -\nabla V.$$

Using

$$\mathbf{D} = \epsilon \mathbf{E},$$

we get

$$\mathbf{D} = -\epsilon \nabla V.$$

Gauss's law says

$$\nabla \cdot \mathbf{D} = \rho.$$

Therefore,

$$\nabla \cdot (-\epsilon \nabla V) = \rho.$$

For constant ϵ ,

$$-\epsilon \nabla^2 V = \rho.$$

Therefore,

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

This is **Poisson's equation**.

It tells you how charge density determines the curvature of voltage.

32 Laplace's Equation

If there is no charge,

$$\rho = 0.$$

Poisson's equation becomes

$$\nabla^2 V = 0$$

This is **Laplace's equation**.

A huge amount of electrostatics involves solving Laplace's equation subject to boundary conditions.

33 Boundary Conditions

When an electromagnetic field crosses a boundary between two materials, the field components must satisfy specific conditions.

For the normal component of $\mathbf{D}$,

$$\hat{\mathbf{n}} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \rho_s$$

where:

- $\hat{\mathbf{n}}$ = unit normal pointing from region 1 to region 2
- $\mathbf{D}_1$ = electric flux density on side 1
- $\mathbf{D}_2$ = electric flux density on side 2
- ρ_s = surface charge density

For the tangential electric field,

$$\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0$$

This says the tangential component of $\mathbf{E}$ is continuous across an ordinary static boundary.

34 Electromagnetic Waves

A simple plane electromagnetic wave can be represented as

$$\mathbf{E} = \hat{\mathbf{x}}E_0 \cos(\omega t - kz)$$

where:

- E_0 = electric-field amplitude
- ω = angular frequency
- t = time
- k = wavenumber
- z = position

The argument

$$\omega t - kz$$

determines the phase of the wave.

The electric field points in the x direction while the wave propagates in the $+z$ direction.

35 Frequency, Period, and Angular Frequency

Frequency is measured in hertz.

$$T = \frac{1}{f}$$

where:

- T = period in seconds
- f = frequency in hertz

Angular frequency is

$$\omega = 2\pi f$$

where:

- ω = angular frequency in radians/second
- f = frequency in hertz

36 Wavelength and Wavenumber

Wavelength is the physical distance corresponding to one complete cycle.

It is represented by

$$\lambda.$$

The wavenumber is

$$k = \frac{2\pi}{\lambda}$$

where:

- k = wavenumber in radians/meter
- λ = wavelength in meters

The factor 2π appears because one complete cycle corresponds to 2π radians.

37 Wave Velocity

Wave velocity is related to frequency and wavelength by

$$v = f\lambda$$

Since

$$\omega = 2\pi f$$

and

$$k = \frac{2\pi}{\lambda},$$

we can also write

$$v = \frac{\omega}{k}$$

The key relationships are

$$T = \frac{1}{f}, \quad \omega = 2\pi f, \quad k = \frac{2\pi}{\lambda}, \quad v = f\lambda, \quad v = \frac{\omega}{k}$$

38 Electromagnetic Wave Velocity

In a homogeneous medium,

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

where:

- v = electromagnetic wave velocity
- μ = permeability
- ϵ = permittivity

In free space,

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

where c is the speed of light.

39 Intrinsic Impedance

The ratio of electric-field magnitude to magnetic-field magnitude in a uniform plane wave is the intrinsic impedance.

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

and

$$\frac{E}{H} = \eta$$

For free space,

$$\eta_0 \approx 377 \, \Omega$$

This is analogous to an impedance relationship between voltage and current, but here it describes the ratio between electric and magnetic field strengths.

40 Poynting Vector

The Poynting vector describes electromagnetic power flow.

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

where:

- $\mathbf{S}$ = power density
- $\mathbf{E}$ = electric field
- $\mathbf{H}$ = magnetic field

The units are

$$\text{W/m}^2.$$

The direction of $\mathbf{S}$ tells you the direction in which electromagnetic energy is flowing.

41 Complex Numbers

Electrical engineering frequently uses complex numbers.

The imaginary unit is

$$j = \sqrt{-1}$$

Electrical engineers use j instead of i because i commonly represents current.

Euler's formula is

$$e^{j\theta} = \cos \theta + j \sin \theta$$

This is extremely important because it connects exponentials with sinusoidal signals.

42 Phasors

Suppose a sinusoidal signal is

$$E(t) = E_0 \cos(\omega t + \phi).$$

We can represent its amplitude and phase using a complex phasor:

$$\tilde{E} = E_0 e^{j\phi}$$

The physical signal is recovered by taking the real part:

$$E(t) = \text{Re} \left\{ \tilde{E} e^{j\omega t} \right\}.$$

One of the major benefits of phasors is that differentiation with respect to time becomes multiplication by $j\omega$:

$$\frac{\partial}{\partial t} \longrightarrow j\omega$$

Therefore,

$$\frac{\partial E}{\partial t} \longrightarrow j\omega \tilde{E}$$

This makes many electromagnetic equations much easier to solve in the frequency domain.

43 Fourier Transform

The Fourier transform converts a time-domain signal into its frequency-domain representation.

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

where:

- $x(t)$ = time-domain signal

- $X(f)$ = frequency-domain representation
- f = frequency
- t = time

The inverse transform is

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df.$$

The big idea is:

time domain $\longleftrightarrow$ frequency domain

44 DFT vs FFT

The Discrete Fourier Transform is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

where:

- $x[n]$ = input samples
- $X[k]$ = frequency-bin output
- N = number of samples
- n = time-domain sample index
- k = frequency-bin index

The DFT is the mathematical operation.

The FFT is an efficient algorithm for computing the DFT.

In simple terms:

FFT = fast algorithm for computing the DFT

For a naive DFT, computational complexity is approximately

$$O(N^2).$$

For a typical FFT algorithm,

$$O(N \log N).$$

This is why FFTs are so useful for SDR.

45 Sampling

When converting an analog signal into digital samples, the sampling rate must be sufficiently high to represent the signal without aliasing.

The Nyquist condition is

$$f_s > 2f_{\max}$$

where:

- f_s = sampling frequency
- $f_{\max}$ = highest frequency component of the signal

If the condition is violated, frequency components can alias into other frequencies.

In SDR, this is one of the fundamental reasons that sample rate and filtering matter.

46 Convolution

Convolution describes how a linear time-invariant system transforms an input signal.

In the time domain,

$$y(t) = x(t) * h(t)$$

where:

- $x(t)$ = input
- $h(t)$ = impulse response
- $y(t)$ = output

In the frequency domain, convolution becomes multiplication:

$$Y(f) = X(f)H(f)$$

This is one of the major reasons Fourier analysis is so useful for filters.

For SDR:

$$\text{filtering in time} \longleftrightarrow \text{multiplication in frequency}$$

47 Divergence Theorem

The divergence theorem connects a volume integral to a closed surface integral.

$$\iiint_V \nabla \cdot \mathbf{A} \, dv = \oint_S \mathbf{A} \cdot d\mathbf{S}$$

In words:

The total amount of divergence inside a volume equals the net outward flux through its enclosing surface.

This theorem is essential for understanding the integral form of Gauss's law.

48 Stokes' Theorem

Stokes' theorem connects curl over a surface to circulation around the boundary.

$$\boxed{\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}}$$

In words:

The total curl passing through a surface equals the circulation around the boundary of that surface.

This theorem is essential for understanding the integral form of Faraday's law and Ampere's law.

49 Differential Equations

A differential equation contains derivatives of an unknown function.

For example,

$$\frac{d^2 y}{dx^2} + ky = 0.$$

A common solution is sinusoidal:

$$y(x) = A \cos(kx) + B \sin(kx).$$

Electromagnetic fields are governed by differential equations because Maxwell's equations describe how fields vary across space and time.

The wave equation is a particularly important example.

50 Separation of Variables

Separation of variables is a technique for solving certain partial differential equations.

For example, assume

$$\boxed{V(x, y) = X(x)Y(y)}$$

where:

- $V(x, y)$ = unknown two-dimensional function
- $X(x)$ = function depending only on x
- $Y(y)$ = function depending only on y

The goal is to turn one complicated PDE into multiple simpler ordinary differential equations.

51 Master Equation Sheet

These are the equations to know especially well.

Vectors

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|}$$

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta$$

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin \theta \hat{\mathbf{n}}$$

Vector Calculus

$$\nabla V = \frac{\partial V}{\partial x} \hat{\mathbf{x}} + \frac{\partial V}{\partial y} \hat{\mathbf{y}} + \frac{\partial V}{\partial z} \hat{\mathbf{z}}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A}$$

$$\nabla^2 V = \nabla \cdot (\nabla V)$$

Electrostatics

$$\mathbf{E} = -\nabla V$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$Q = \int_V \rho \, dv$$

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

$$\nabla^2 V = 0 \quad (\rho = 0)$$

Magnetics

$$\mathbf{B} = \mu \mathbf{H}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

Maxwell

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Waves

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

$$v = f\lambda$$

$$v = \frac{\omega}{k}$$

$$k = \frac{2\pi}{\lambda}$$

$$\omega = 2\pi f$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Complex Numbers and Phasors

$$j = \sqrt{-1}$$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$\frac{\partial}{\partial t} \longrightarrow j\omega$$

Fourier and SDR

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

$$f_s > 2f_{\max}$$

$$Y(f) = X(f)H(f)$$

Vector Integral Theorems

$$\iiint_V \nabla \cdot \mathbf{A} dv = \oint_S \mathbf{A} \cdot d\mathbf{S}$$

$$\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$

52 Symbol Reference

Symbol	Meaning	Typical Unit
E	Electric field	V/m
D	Electric flux density	C/m ²
B	Magnetic flux density	T
H	Magnetic field intensity	A/m
J	Current density	A/m ²
<i>V</i>	Electric potential	V
<i>Q</i>	Electric charge	C
ρ	Volume charge density	C/m ³
ρ_s	Surface charge density	C/m ²
ρ_l	Line charge density	C/m
ϵ	Permittivity	F/m
μ	Permeability	H/m
σ	Conductivity	S/m
η	Intrinsic impedance	Ω
<i>f</i>	Frequency	Hz
<i>T</i>	Period	s
ω	Angular frequency	rad/s
λ	Wavelength	m
<i>k</i>	Wavenumber	rad/m
<i>c</i>	Speed of light	m/s
S	Poynting vector	W/m ²

53 What to Memorize First

Do not try to memorize everything simultaneously.
Start with these.

1. Vector Calculus

Know what these mean:

$$\boxed{\nabla V}$$

$$\boxed{\nabla \cdot \mathbf{A}}$$

$$\boxed{\nabla \times \mathbf{A}}$$

$$\boxed{\nabla^2 V}$$

Remember:

$$\boxed{\text{gradient} \rightarrow \text{scalar to vector}}$$

$$\boxed{\text{divergence} \rightarrow \text{vector to scalar}}$$

$$\boxed{\text{curl} \rightarrow \text{vector to vector}}$$

2. Electric Field and Voltage

$$\boxed{\mathbf{E} = -\nabla V}$$

3. Constitutive Relations

$$\boxed{\mathbf{D} = \epsilon \mathbf{E}}$$

$$\boxed{\mathbf{B} = \mu \mathbf{H}}$$

$$\boxed{\mathbf{J} = \sigma \mathbf{E}}$$

4. Maxwell's Equations

Memorize all four.

5. Wave Relationships

$$v = f\lambda$$

$$\omega = 2\pi f$$

$$k = \frac{2\pi}{\lambda}$$

$$v = \frac{\omega}{k}$$

6. Fourier Analysis

Understand the idea that

$$\text{time domain} \longleftrightarrow \text{frequency domain}$$

and know the DFT/FFT relationship.

54 The Most Important Mental Map

Try to understand electromagnetics as one connected system.

Step 1: Charge

Charge density creates electric flux:

$$\nabla \cdot \mathbf{D} = \rho$$

Step 2: Electric Field

Electric field is related to voltage:

$$\mathbf{E} = -\nabla V$$

Step 3: Material

The material determines how $\mathbf{D}$ relates to $\mathbf{E}$:

$$\mathbf{D} = \epsilon \mathbf{E}$$

Step 4: Current

Conducting materials allow current:

$$\mathbf{J} = \sigma \mathbf{E}$$

Step 5: Magnetic Field

Magnetic fields are related to currents and changing electric fields:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Step 6: Changing Magnetic Fields

Changing magnetic fields create electric fields:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Step 7: Waves

The electric and magnetic fields can therefore continually generate one another.

That produces electromagnetic waves.

$$\text{changing } \mathbf{E} \rightarrow \text{changing } \mathbf{H} \rightarrow \text{changing } \mathbf{E} \rightarrow \dots$$

55 Maxwell's Equations as a Mental Map

Instead of memorizing Maxwell's equations as four unrelated equations, associate each with a question.

Question 1: What creates electric flux?

$$\nabla \cdot \mathbf{D} = \rho$$

Answer:

electric charge.

Question 2: What creates magnetic flux divergence?

$$\nabla \cdot \mathbf{B} = 0$$

Answer:

there are no isolated magnetic monopoles.

Question 3: What creates circulating electric fields?

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Answer:

changing magnetic fields.

Question 4: What creates circulating magnetic fields?

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Answer:

electric current and changing electric fields.

56 Three Questions to Ask for Every Equation

When looking at a new electromagnetics equation, ask:

1. What goes into the equation?

For example,

$$\nabla \cdot \mathbf{D}$$

takes a vector field and produces a scalar.

2. What comes out?

For example,

$$\nabla \cdot \mathbf{D} = \rho$$

produces charge density.

3. What does it physically mean?

In this case:

$$\text{electric charge is a source of electric flux}$$

This method is often more useful than trying to memorize symbols without understanding them.

57 Final Study Strategy

A useful order for learning the material is:

1. Scalars vs vectors
2. Unit vectors
3. Vector magnitude
4. Dot product
5. Cross product
6. Partial derivatives

7. Gradient
8. Divergence
9. Curl
10. Laplacian
11. Line/surface/volume integrals
12. Flux
13. Charge density
14. Electric potential
15. Electric field
16. Permittivity
17. $\mathbf{D}$
18. $\mathbf{B}$ and $\mathbf{H}$
19. Current density
20. Maxwell's equations
21. Poisson and Laplace equations
22. Boundary conditions
23. Waves
24. Phasors
25. Fourier transforms
26. DFT/FFT
27. Sampling
28. Convolution

For each equation, learn three things:

What does it say?

What does every symbol mean?

What does it mean physically?

Do not worry about memorizing every derivation immediately.
The goal is first to understand the structure.

58 Quick Reference

Vector Operations

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta$$

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin \theta \hat{\mathbf{n}}$$

$$\nabla V = \text{gradient}$$

$$\nabla \cdot \mathbf{A} = \text{divergence}$$

$$\nabla \times \mathbf{A} = \text{curl}$$

$$\nabla^2 V = \text{Laplacian}$$

Electrostatics

$$\mathbf{E} = -\nabla V$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

$$\nabla^2 V = 0 \quad \text{if } \rho = 0$$

Magnetics

$$\mathbf{B} = \mu \mathbf{H}$$

$$\nabla \cdot \mathbf{B} = 0$$

Maxwell

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Waves

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

$$v = f\lambda$$

$$v = \frac{\omega}{k}$$

$$\omega = 2\pi f$$

$$k = \frac{2\pi}{\lambda}$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Fourier / SDR

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$$

$$f_s > 2f_{\max}$$

$$Y(f) = X(f)H(f)$$

59 Core Mental Model

If everything above feels overwhelming, remember this simplified model:

Charge $\rightarrow$ Electric Field

Current $\rightarrow$ Magnetic Field

Changing Magnetic Field $\rightarrow$ Electric Field

Changing Electric Field $\rightarrow$ Magnetic Field

Therefore:

Electric and magnetic fields can propagate together as waves.

For a plane wave:

$\mathbf{E} \perp \mathbf{H} \perp$ direction of propagation

And the propagation velocity is

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

For free space,

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

Finally, for SDR and signal processing:

RF signal $\rightarrow$ sampling $\rightarrow$ IQ data $\rightarrow$ Fourier/filtering $\rightarrow$ information

That is the bridge between the electromagnetics you're learning and the RF/SDR work you're already doing.